SUTRA WORKFLOW STATUS REPORT

Project update and test evidence | 11 October 2026

EXECUTIVE CONCLUSION

The repository now has a rail-first deterministic recovery slice and a new versioned API boundary. The canonical fixture is Mang

The implementation is not yet fully test-gated. Frontend dependency installation failed in the environment, so TypeScript, ESLin

WORK COMPLETED

W03-W08 Domain foundation - Flight demo data replaced with rail-first fixtures; delay/cancellation, impact relationships, candida

W09 API boundary - Added GET /api/v1/trips/:tripId and POST /api/v1/recovery-runs with status, impact, candidates, recommen

W19-W22 AI boundary - Recovery engine exposes an AI-shaped verified-candidate decision and deterministic fallback mode; in

W10-W11 UI domain wording - Disruption view now presents Rail rather than Flight while preserving the journey, impact, compa

Regression hygiene - git diff --check passed; no new tracked cache or generated dependency directory remains.

TEST CASES AND RESULTS

T01 PASS - Rail fixture contains RAIL-A, RAIL-B, HOTEL-1. Static source assertion passed.

T02 PASS - Canonical recovery route exposes NO_FEASIBLE_RECOVERY. Static source assertion passed.

T03 PASS - Trip route exposes not-found semantics. Static source assertion passed.

T04 PASS - Engine contains deterministic fallback boundary. Static source assertion passed.

T05 PASS - Patch whitespace and merge hygiene. git diff --check returned clean.

T06 BLOCKED - TypeScript typecheck. npm run typecheck could not find tsc because node_modules is absent.

T07 BLOCKED - Dependency installation. npm ci failed with npm EPERM cache access, then failed with npm internal exit-hand

T08 NOT RUN - ESLint requires installed frontend dependencies.

T09 NOT RUN - Production build requires installed frontend dependencies and font/build runtime.

T10 NOT RUN - Browser E2E, responsive, accessibility, and reduced motion require a successful build and browser dependenc

PLANNING WORKFLOW COVERAGE

M1 Contract Freeze - PARTIAL. Canonical rail fixture and v1 API routes are present; frontend client and hook still use transition

M2 Golden Vertical Slice - PARTIAL. Rail fixture, impact, candidates, constraints, engine, and UI wording are present; integrate

M3 Reliability Core - PARTIAL. Cancellation, no-feasible, and technical-error scenarios remain in the prototype; v1 technical-err

M4 AI Layer - PARTIAL. Verified-feasible selection and fallback boundary are represented; no external AI provider is connected

M5 QA Gate - NOT COMPLETE. Adversarial, data-integrity, AI-output, and four-scenario regression tests still need executable

M6-M7 Demo and submission - NOT COMPLETE. Build/deployment validation, demo freeze, evidence set, PPT, and recording

REMAINING IMPLEMENTATION WORK

1. Move the frontend client and recovery hook from transitional scenario endpoints to the authoritative v1 recovery-run flow.

2. Add executable unit tests for impact propagation, candidate feasibility, original-itinerary preservation, cancellation, infeasibility

3. Install dependencies successfully and rerun typecheck, lint, build, browser E2E, responsive, accessibility, and reduced-motio

4. Resolve compile or runtime findings, then complete the four-scenario regression and deployment smoke test.

5. Finish requirements traceability, judge-facing UX review, demo freeze, and submission evidence.

FILES CHANGED

frontend/lib/types.ts

frontend/lib/mock.ts

frontend/lib/engine.ts

frontend/components/sutra/sections.tsx

frontend/app/api/v1/trips/[tripId]/route.ts

frontend/app/api/v1/recovery-runs/route.ts

VERIFICATION STATEMENT

Implementation claims are limited to the source-level changes and commands recorded above. A green product status requires